

★ PRIMER · UNIT 16

Before the Lecture: The Whole Pipeline

Unit 16 · Integration and Capstone

Watch this first before you scope your project

Twenty minutes of groundwork — the five stages of a finished project, what a costed return looks like, and the five ways good projects quietly fail.

How to Use This Primer

Two audiences, one deck — read the column that describes you

If this is new to you

- Watch before you scope.
- The five stages are here.
- Do slide 8's arithmetic.

If you have met it before

- Skim it — this is revision.
- Slide 7 is what finance changes.
- Slide 16 says if you can skip.

Nothing here is examined on its own — it is the floor the lecture stands on

Where Unit 16 Sits

The course builds one pipeline; here is this unit's stone

Unit 15

done

Transformers, graphs and agents.

Unit 16

you are here

Everything, assembled into one pipeline.

Week 16

exam

The final examination — units 9 to 16.

Nothing new is taught here. What is taught is how the fifteen previous units fit together.

A Project Is Five Stages, Not One Model

Most failed projects are excellent models attached to a broken stage either side of them.

Where attention goes

- Choosing the algorithm.
- Tuning the parameters.
- Perhaps 10% of the value.

Where the value is

- The label, and the data.
- Honest testing.
- Costs, and the write-up.

Your grade follows the same distribution. The model choice is the smallest part of it.

The Five Stages

Every stage maps to a unit you have already done

Stage	What happens	Unit
Data	bars, cleaning, features	2
Label	what exactly you predict	3
Model	fit something, honestly	4 to 9
Validate	purged CV, deflated Sharpe	10
Decide	size it, cost it, explain it	11 and 14

If you cannot name the unit behind a stage, that is the stage to go back and shore up.

The Five Most Common Project Mistakes

Each one is a unit's warning, ignored

Mistake	What it looks like	Warned in
No costed return	a beautiful gross Sharpe	unit 10
Accuracy on rare events	97% and useless	unit 3
Leakage	suspiciously good results	unit 10
One split, one number	no sense of the spread	unit 10
No baseline	nothing to compare against	unit 11

Every one of these makes results look better. That is why they survive to the presentation.

Costs, Before the Applause

A strategy that turns over daily needs a large gross edge just to arrive at zero after costs.

What gets forgotten

- Commission per trade.
- The bid-ask spread.
- Market impact when you size up.

What it does

- Halves a typical gross edge.
- Punishes fast turnover most.
- Decides the whole verdict.

Report gross and net side by side. A project that reports only gross has not finished.

The Accuracy Trap, Priced

A fraud model on 1,000 transactions, 30 of them fraudulent

Model	Caught	Missed	Accuracy	Useful?
Predict 'never fraud'	0	30	97.0%	no
A real model	24	6	96.2%	yes
Difference	+24	-24	-0.8pp	everything

The useful model scores lower on accuracy. Report precision and recall, or report nothing.

Net of Costs, by Hand

The same strategy, before and after the friction

Gross

- 12% a year.
- Sharpe 1.4.
- Presentation-ready.

Net

- 40 round trips, 6bp each.
- Costs: 2.4% a year.
- 9.6% and Sharpe 1.1.

Turnover is the multiplier. Halve the trading and most of that cost disappears

The Luck Bar

How good must your result be, given how many things you tried?

One honest attempt

- Sharpe 1.0 is decent.
- Hard to reach by chance.
- Worth reporting.

Two hundred attempts

- The best will exceed 1.0.
- By luck alone.
- Report the count.

Say how many configurations you tried. Examiners trust a stated number far more than silence.

How a Great Project Is Judged

Three questions, and the model is only in the first

1 · Is it sound?

the pipeline

Honest labels, purged
validation, no leakage
anywhere.

2 · Is it real?

the evidence

Net of costs, against a baseline,
with the attempt count stated.

3 · Is it clear?

the write-up

Someone else can rerun it and
reach the same number.

A modest result, honestly established, beats a spectacular one you cannot defend.

Your Toolkit, by Job

When you next face a problem, this is the table to open

The job	Reach for	Unit
Predict a category	logistic, trees, XGBoost	5, 7
Find structure	k-means, PCA, denoising	9
Build a portfolio	HRP on a clean matrix	9, 11
Read documents	embeddings and RAG	12
Explain a decision	SHAP	14

Sixteen units, five jobs. Most real work is one of these five, or two of them joined.

Notation You Will Meet

Four things to state explicitly in your report

State this	Why	Where
Attempt count	the luck bar	validation section
Net of costs	gross means nothing	results section
The baseline	beat $1/N$ or a naive rule	results section
The split dates	so it can be rerun	methods section

Four lines. Their absence is what separates a good project from a defensible one.

The Python You Will See

You do not need to write it yet — you need to recognise it

Line	What it does
<code>np.random.seed(42)</code>	so your numbers reproduce
<code>pipe = Pipeline([...])</code>	scaling inside the fold, not before
<code>cross_val_score(pipe, X, y, cv=purged)</code>	the honest score
<code>ret - turnover * cost_bps / 1e4</code>	net of costs, in one line
<code>pd.DataFrame(results).to_csv()</code>	the scoreboard your report quotes

Set the seed on line one. A project that cannot reproduce its own number cannot be marked.

Three Things Beginners Get Wrong

Each one costs marks, not just time

1

"I will pick the model first"

Scope the label and the validation first. The model is the easiest decision you will make.

2

"I will add costs at the end if there is time"

Costs change which strategy wins. Added at the end they change your conclusion, not your appendix.

3

"A negative result is a failed project"

An honestly established negative result is a complete project. A hidden one is a failed career.

If one of these surprised you, that is exactly what the primer was for

Self-Check Before the Lecture

Answer these three out loud — if you can, you are ready

1	1,000 rows, 30 frauds. Accuracy of 'never fraud'?	97.0%
2	40 round trips at 6bp. What is the annual cost?	2.4%
3	Which four things must the report state?	see slide 13

Cover the right column, answer, then check. All three come from this deck.

What the Lecture Does With This

You bring the vocabulary; the lecture brings the argument

1

It puts sixteen units on one page

The whole journey, the toolkit by job, and the lessons bigger than any algorithm.

2

It sets the standard for the project

Five stages, the five common mistakes, the luck bar and the cost line, all priced.

3

It closes the course

Financial machine learning is not a model. It is a pipeline, and you have now built every stage.

Next: Unit 16 lecture — Bringing It All Together